



SEN 432: DISTRIBUTED AND CLOUD COMPUTING

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CLOUD MONITORING & ANALYTICS

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Learning objectives

- In this module, you will learn how to:
 - Summarize approaches to monitoring your AWS environment.
 - Describe the benefits of Amazon CloudWatch.
 - Describe the benefits of AWS CloudTrail.
 - Describe the benefits of AWS Trusted Advisor.

Introduction

- Every business can use metrics to measure how well systems and processes are running.
- This idea of observing systems, collecting metrics, evaluating those metrics over time, and then using them to make decisions or take actions, is called **monitoring**. It's important to set up monitoring in the cloud.
- With the elastic nature of AWS services that dynamically scale up and down, you'll want to keep a close pulse on your AWS resources to ensure that your systems are running as expected.
- **For example**, if an EC2 instance is being over-utilized, you can trigger a scaling event that automatically would launch another EC2 instance.
- Or if an application starts sending error responses at an unusually high rate, you can alert an employee to go look at what's going on.

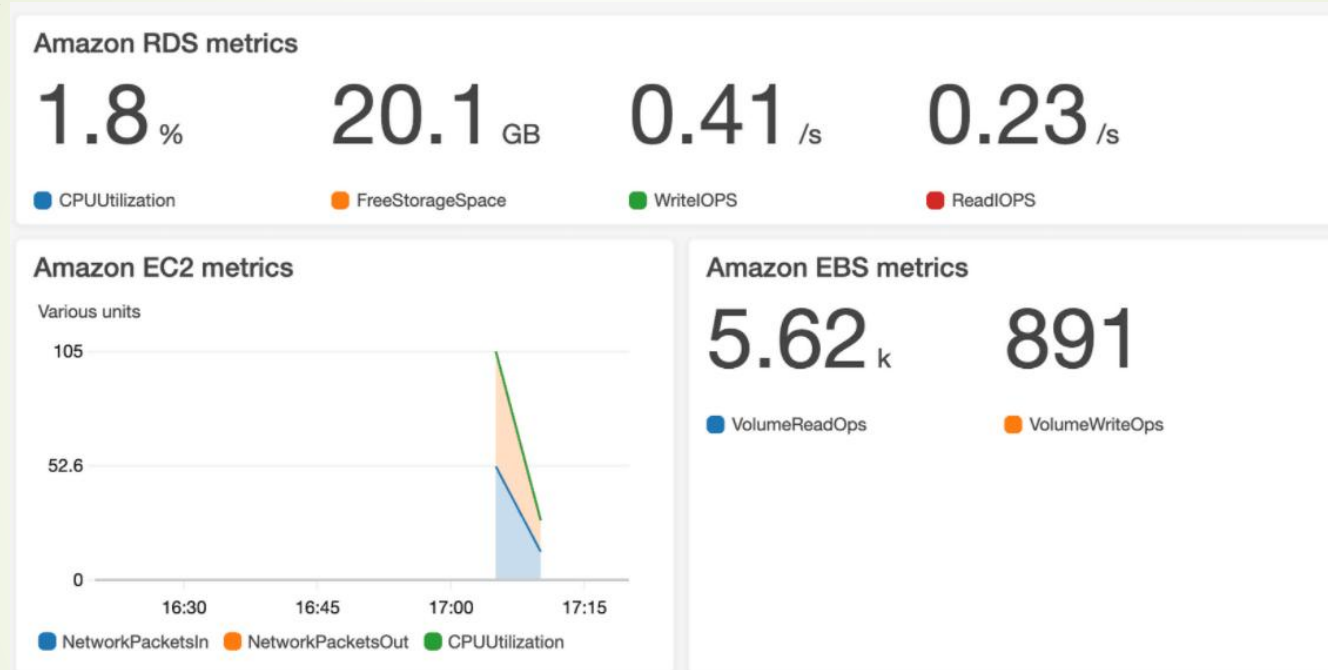
Amazon CloudWatch

- Amazon CloudWatch is a web service that enables you to **monitor and manage** various metrics and **configure alarm** actions based on data from those metrics.
- CloudWatch uses metrics to represent the data points for your resources.
- Think of metrics as variables that are tied to your resources.
- AWS services send metrics to CloudWatch. CloudWatch then uses these metrics to create graphs automatically that show how performance has changed over time.
- **CloudWatch alarms:** With CloudWatch, you can create alarms that automatically perform actions if the value of your metric has gone above or below a predefined threshold.

Amazon CloudWatch Alarm

- For example, suppose that your company's developers use Amazon EC2 instances for application development or testing purposes. If the developers occasionally forget to stop the instances, the instances will continue to run and incur charges.
- In this scenario, you could **create a CloudWatch alarm that automatically stops an Amazon EC2 instance when the CPU utilization percentage has remained below a certain threshold for a specified period.**
- When configuring the alarm, you can specify to receive a notification whenever this alarm is triggered.
- CloudWatch alarms are integrated with **SNS**. So, it can send an SMS to the manager.
- You can create all sorts of custom alarms for metrics from all different types of AWS resources.

CloudWatch dashboard



- The CloudWatch [dashboard](#) feature enables you to access all the metrics for your resources from a single location. For example, you can use a CloudWatch dashboard to monitor the CPU utilization of an Amazon EC2 instance, the total number of requests made to an Amazon S3 bucket, and more. You can even customize separate dashboards for different business purposes, applications, or resources.

Amazon CloudWatch - Benefits

- Provides **access to all your metrics from a central location**. This enables you to collect metrics and logs from all your AWS resources applications, and services that run on AWS and on-premises servers, helping you break down silos so that you can easily gain system-wide visibility.
- You can also get **visibility across your applications, infrastructure, and services**, which means you gain insights across your distributed stack so you can correlate and visualize metrics and logs to quickly pinpoint and resolve issues.
 - This in turn means you can reduce **Mean Time To Resolution, or MTTR**, and improve **total cost of ownership, or TCO**. This means freeing up important resources like developers to focus on adding business value.
- Lastly, you can **drive insights to optimize applications and operational resources**. By, for example, aggregating usage across an entire fleet of EC2 instances to derive operational and utilization insights.

AWS CloudTrail

- ▶ **AWS CloudTrail** records API calls for your account. The recorded information includes the **identity of the API caller, the time of the API call, the source IP address of the API caller,** and more. You can think of CloudTrail as a “trail” of breadcrumbs (or a log of actions) that someone has left behind them.
- ▶ Recall that you can use API calls to provision, manage, and configure your AWS resources. With CloudTrail, you can view a complete history of user activity and API calls for your applications and resources.
- ▶ Events are typically **updated in CloudTrail within 15 minutes after an API call**. You can **filter events** by specifying the time and date that an API call occurred, the user who requested the action, the type of resource that was involved in the API call, and more.

Example: AWS CloudTrail event

- Suppose that the account owner is browsing through the AWS Identity and Access Management (IAM) section of the AWS Management Console and discovers that a new IAM user named Mary was created, but does not know who, when, or which method created the user. To answer these questions, the owner navigates to AWS CloudTrail.

<u>What</u> happened?	A new IAM user (Mary) was created.	
<u>Who</u> made the request?	IAM user John	
<u>When</u> did this occur?	January 1, 2020 at 9:00 AM	
<u>How</u> was the request made?	Through the AWS Management Console	

CloudTrail Insights

- Within CloudTrail, you can also enable [CloudTrail Insights](#).
- This optional feature allows CloudTrail to **automatically detect unusual API activities in your AWS account**.
- For example, CloudTrail Insights might detect that a higher number of Amazon EC2 instances than usual have recently launched in your account. You can then review the full event details to determine which actions you need to take next.

Knowledge check

- Which tasks can you perform using AWS CloudTrail? (Select TWO.)
- A. Monitor your AWS infrastructure and resources in real time
- B. Track user activities and API requests throughout your AWS infrastructure
- C. View metrics and graphs to monitor the performance of resources
- D. Filter logs to assist with operational analysis and troubleshooting
- E. Configure automatic actions and alerts in response to metrics
- B, D
- The other response options are tasks that you can perform in Amazon CloudWatch.

AWS Trusted Advisor

- AWS Trusted Advisor is a web service that inspects your AWS environment and provides real-time recommendations in accordance with AWS best practices.
- Trusted Advisor compares its findings to AWS best practices in five categories: ***cost optimization, performance, security, fault tolerance, and service limits***.
- For the checks in each category, Trusted Advisor offers a list of recommended actions and additional resources to learn more about AWS best practices.
- The guidance provided by AWS Trusted Advisor can benefit a company at all stages of deployment.
- For example, you can use AWS Trusted Advisor to assist you while you are creating new workflows and developing new applications. Or you can use it while you are making ongoing improvements to existing applications and resources.

AWS Trusted Advisor dashboard

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- When you access the Trusted Advisor dashboard on the AWS Management Console, you can review completed checks for cost optimization, performance, security, fault tolerance, and service limits.
- For each category:
 - The green check indicates the number of items for which it detected no problems.
 - The orange triangle represents the number of recommended investigations.
 - The red circle represents the number of recommended actions.
- Some checks are free and are included in your AWS account, and others are available depending on the level of your support plan.
- Some examples of checks are, if you don't have multi-factor authentication turned on for your root user, it's going to let you know.

Examples of Issues : Cost Optimization & Performance

- If you have **underutilized EC2 instances** that might be able to be turned off in order to save money, or if you have **EBS volumes that haven't been backed up** in a reasonable amount of time, it will let you know that, too.
- If you have some **RDS instances that are idle**, as well as some underutilized EC2 instances and EBS volumes.
- You could decide to scale these instances down vertically to save on cost, or if they aren't being used at all you could decide to delete those instances and those EBS volumes altogether.
- **Performance:** checks for EBS volumes whose performance might've been affected by the throughput capability of the EC2 instance that it's attached to.

Examples of Issues : Security, Fault Tolerance & Service Limits

- **Security issues:** weak password policies for IAM users, multi-factor authentication is not turned on for the root user, and there are security groups allowing public access to EC2 instances. All of these items are putting the resources in this account at risk, and should be dealt with as soon as possible.
- **Fault Tolerance:** EBS volumes without snapshots. Remember, a snapshot is a backup. So, without backups, if I had an EBS volume fail, data would be lost.
- If you have EC2 instances are not properly launched across AZs. So, if one AZ has trouble, your application might incur an outage. Deploying across AZs would be the answer to this one.
- **Service Limits:** An account having five VPCs for example, which is the regional limit. That is a specific service limit.

Service Limits

- This pillar will tell you when you are approaching or hitting AWS service limits.
- A lot of service limits are soft limits, meaning they are restrictions that can be lifted to some degree.
- It's good to know when you are approaching one of those limits, and when it's time to turn in a Support ticket with AWS to get them changed.
- Trusted Advisor can help point you in the right direction when it comes to the five pillars. You can set up email alerts that go out to billing, operations, and security contacts, as checks get run in your account.

Additional resources

- To learn more about the concepts that were explored in this Module, review these resources.
- [Management and Governance on AWS](#)
- [Monitoring and Observability](#)
- [Configuration, Compliance, and Auditing](#)
- [AWS Management & Governance Blog](#)
- [Whitepaper: AWS Governance at Scale](#)

CLOUD PRICING AND SUPPORT

Prepared by: Dr. Bilkisu L. Muhammad-Bello

Learning objectives

- In this module, you will learn how to:
 - Describe AWS pricing and support models.
 - Describe the AWS Free Tier.
 - Describe key benefits of AWS Organizations and consolidated billing.
 - Explain the benefits of AWS Budgets.
 - Explain the benefits of AWS Cost Explorer.
 - Explain the primary benefits of the AWS Pricing Calculator.
 - Distinguish between the various AWS Support Plans.
 - Describe the benefits of AWS Marketplace.

Introduction

- AWS provides a lot of free tools to help you:
 - Try out some services for free
 - Breakdown and visualize the different aspects of your cloud expenditure
 - Estimate your expenditure
 - Plan and analyze your budgets for your AWS environments.

AWS Free Tier

- The [AWS Free Tier](#) enables you to begin using certain services without having to worry about incurring costs for the specified period.
- Three types of offers are available:
 - Always Free
 - 12 Months Free
 - Trials
- For each free tier offer, make sure to review the specific details about exactly which resource types are included.

Always Free

- These offers do not expire and are available to all AWS customers.
- Let's take AWS Lambda, our serverless compute option. **As of March 2020, it allows for one million free invocations per month.** That means if you stay under one million invocations, it's always free.
- AWS Lambda also allows up to 3.2 million seconds of compute time per month.
- Amazon DynamoDB allows 25 GB of free storage per month.
- Some other services that qualify under the free tier are: SageMaker, Comprehend Medical, DynamoDB, SNS, Cognito, and so much more.

12 Months Free

- These offers are free for 12 months following your initial sign-up date to AWS.
- Examples include specific amounts of Amazon S3 Standard Storage, thresholds for monthly hours of Amazon EC2 compute time, and amounts of Amazon CloudFront data transfer out.
- S3 which is an object store service is free for 12 months for up to 5GBs of storage.
- Thereafter, you'll incur a cost. Great for trying out a static website on S3, I might say.

Trials

- ▶ Short-term free trial offers start from the date you activate a particular service. The length of each trial might vary by number of days or the amount of usage in the service.
- ▶ For example, **Amazon Inspector** offers a 90-day free trial.
- ▶ Amazon Lightsail (a service that enables you to run virtual private servers) offers 750 free hours of usage over a 30-day period.
- ▶ **Lightsail** is a service where you can deploy ready-made application stacks. The one-month free trial offer of up to 750 hours of usage.
- ▶ That's a pretty decent amount to deploy and test a WordPress blog.

Knowledge check

- The AWS Free Tier includes offers that are available to new AWS customers for a certain period of time following their AWS sign-up date. What is the duration of this period?
- A. 3 months
 - B. 6 months
 - C. 9 months
 - D. 12 months
- D

AWS pricing concepts

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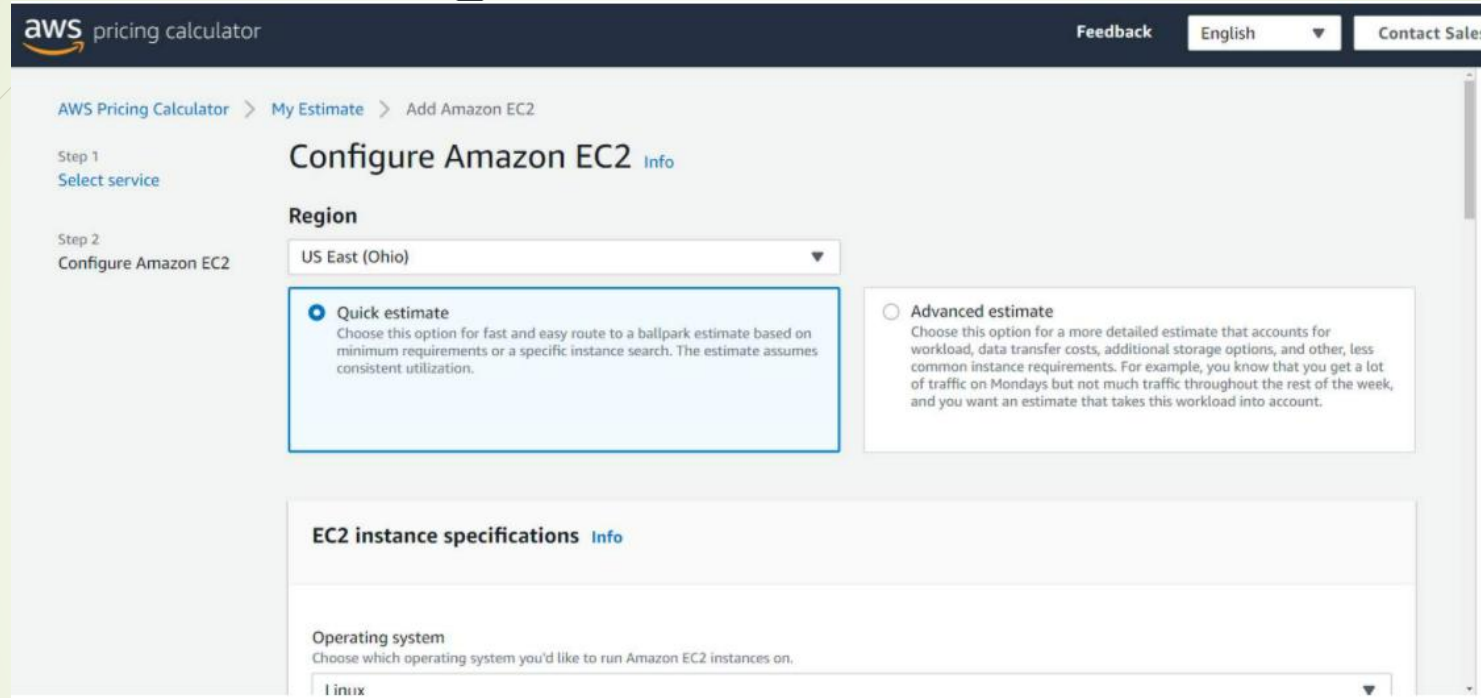
How AWS pricing works

- AWS offers a range of cloud computing services with pay-as-you-go pricing.
- **Pay for what you use:** For each service, you pay for exactly the amount of resources that you actually use, without requiring long-term contracts or complex licensing.
- **Pay less when you reserve:** Some services offer reservation options that provide a significant discount compared to On-Demand Instance pricing.
 - For example, suppose that your company is using Amazon EC2 instances for a workload that needs to run continuously. You might choose to run this workload on Amazon EC2 Instance Savings Plans, because the plan allows you to save up to 72% over the equivalent On-Demand Instance capacity.
- **Pay less with volume-based discounts when you use more:** Some services offer tiered pricing, so the per-unit cost is incrementally lower with increased usage.
 - For example, the more Amazon S3 storage space you use, the less you pay for it per GB.

AWS Pricing Calculator

- The [AWS Pricing Calculator](#) lets you explore AWS services and create an estimate for the cost of your use cases on AWS.
- You can organize your AWS estimates by groups that you define. A group can reflect how your company is organized, such as providing estimates by cost center.
- When you have created an estimate, you can save it and generate a link to share it with others.

AWS Pricing Calculator



The screenshot shows the AWS Pricing Calculator interface. At the top, there's a dark blue header with the AWS logo, 'pricing calculator', and links for 'Feedback', 'English', and 'Contact Sales'. Below the header, a breadcrumb trail reads 'AWS Pricing Calculator > My Estimate > Add Amazon EC2'. The main content area is titled 'Configure Amazon EC2' with an 'Info' link. On the left, a sidebar shows 'Step 1: Select service' and 'Step 2: Configure Amazon EC2'. The 'Region' dropdown is set to 'US East (Ohio)'. There are two radio buttons for 'Quick estimate' (selected) and 'Advanced estimate'. The 'Quick estimate' section explains it's for a ballpark estimate based on minimum requirements. The 'Advanced estimate' section explains it's for a more detailed estimate accounting for workload, data transfer costs, and storage options. Below this, the 'EC2 instance specifications' section is visible, with a dropdown for 'Operating system' set to 'Linux'.

- Suppose that your company is interested in using Amazon EC2. However, you are not yet sure which AWS Region or instance type would be the most cost-efficient for your use case. In the AWS Pricing Calculator, you can enter details such as the kind of operating system you need, memory requirements, and input/output (I/O) requirements. By using the AWS Pricing Calculator, you can review an estimated comparison of different EC2 instance types across AWS Regions.

AWS pricing examples - Lambda

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- For AWS Lambda, you are charged based on the number of requests for your functions and the time that it takes for them to run.
- AWS Lambda allows **1 million free requests** and up to **3.2 million seconds of compute** time per month.
- You can save on AWS Lambda costs by signing up for Compute Savings Plans.
- Compute Savings Plans offer lower compute costs in exchange for committing to a consistent amount of usage over a 1-year or 3-year term. This is an example of **paying less when you reserve**.
- If you have used AWS Lambda in multiple AWS Regions, you can view the itemized charges **by Region** on your bill.
- In this example, all the AWS Lambda usage occurred in the Northern Virginia Region. The bill lists separate charges for the number of requests for functions and their duration.
- Both the number of requests and the total duration of requests in this example are under the thresholds in the AWS Free Tier, so the account owner would not have to pay for any AWS Lambda usage in this month.

▼ Lambda		\$0.00
▼ US East (N. Virginia)		\$0.00
AWS Lambda Lambda-GB-Second		\$0.00
AWS Lambda - Compute Free Tier - 400,000 GB-Seconds - US East (Northern Virginia)	254.575 seconds	\$0.00
AWS Lambda Request		\$0.00
AWS Lambda - Requests Free Tier - 1,000,000 Requests - US East (Northern Virginia)	680.000 Requests	\$0.00

AWS pricing examples – EC2

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- With Amazon EC2, you pay for only the compute time that you use while your instances are running.
- For some workloads, you can significantly reduce Amazon EC2 costs by using **Spot Instances**.
- For example, suppose that you are running a batch processing job that is able to withstand interruptions. Using a Spot Instance would provide you with up to 90% cost savings while still meeting the availability requirements of your workload.
- You can find additional cost savings for Amazon EC2 by considering **Savings Plans** and **Reserved Instances**.

- The service charges in this example include details for the following items:
 - Each Amazon EC2 instance type that has been used
 - The amount of Amazon EBS storage space that has been provisioned
 - The length of time that Elastic Load Balancing has been used
- In this example, all the usage amounts are under the thresholds in the AWS Free Tier, so the account owner would not have to pay for any Amazon EC2 usage in this month.

▼ Elastic Compute Cloud		\$0.00
▼ US East (N. Virginia)		\$0.00
Amazon Elastic Compute Cloud running Linux/UNIX		\$0.00
\$0.00 per Linux t2.micro instance-hour (or partial hour) under monthly free tier	106.512 Hrs	\$0.00
EBS		\$0.00
\$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier	11.294 GB-Mo	\$0.00
Elastic Load Balancing - Application		\$0.00
\$0.00 per Application LoadBalancer-hour (or partial hour) under monthly free tier	268.000 Hrs	\$0.00

AWS pricing examples – S3

- For Amazon S3 pricing, consider the following cost components:
- **Storage** - You pay for only the storage that you use. You are charged the rate to store objects in your Amazon S3 buckets based on your objects' sizes, storage classes, and how long you have stored each object during the month.
- **Requests and data retrievals** - You pay for requests made to your Amazon S3 objects and buckets. For example, suppose that you are storing photo files in Amazon S3 buckets and hosting them on a website. Every time a visitor requests the website that includes these photo files, this counts towards requests you must pay for.
- **Data transfer** - There is no cost to transfer data between different Amazon S3 buckets or from Amazon S3 to other services within the same AWS Region. However, you pay for data that you transfer into and out of Amazon S3, with a few exceptions. There is no cost for data transferred into Amazon S3 from the internet or out to Amazon CloudFront. There is also no cost for data transferred out to an Amazon EC2 instance in the same AWS Region as the Amazon S3 bucket.
- **Management and replication** - You pay for the storage management features that you have enabled on your account's Amazon S3 buckets. These features include Amazon S3 inventory, analytics, and object tagging.

AWS pricing examples – S3

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- The AWS account in this example has used Amazon S3 in two Regions: Northern Virginia and Ohio. For each Region, itemized charges are based on the following factors:
 - The number of requests to add or copy objects into a bucket
 - The number of requests to retrieve objects from a bucket
 - The amount of storage space used
- All the usage for Amazon S3 in this example is under the AWS Free Tier limits, so the account owner would not have to pay for any Amazon S3 usage in this month.

▼ Simple Storage Service		\$0.00
▼ US East (N. Virginia)		\$0.00
Amazon Simple Storage Service Requests-Tier1		\$0.00
\$0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier	185.000 Requests	\$0.00
Amazon Simple Storage Service Requests-Tier2		\$0.00
\$0.00 per request - GET and all other requests under the monthly global free tier	923.000 Requests	\$0.00
Amazon Simple Storage Service TimedStorage-ByteHrs		\$0.00
\$0.000 per GB - storage under the monthly global free tier	0.159 GB-Mo	\$0.00
▼ US East (Ohio)		\$0.00
Amazon Simple Storage Service USE2-Requests-Tier2		\$0.00
\$0.00 per request - GET and all other requests under the monthly global free tier	4.000 Requests	\$0.00
Amazon Simple Storage Service USE2-TimedStorage-ByteHrs		\$0.00
\$0.000 per GB - storage under the monthly global free tier	0.000001 GB-Mo	\$0.00

Billing dashboard

- Use the [AWS Billing & Cost Management dashboard](#) to pay your AWS bill, monitor your usage, and analyze and control your costs.
- 1. Compare your current month-to-date balance with the previous month and get a **forecast** of the next month based on current usage.
- 2. View month-to-date spend by service.
- 3. View Free Tier usage by service.
- 4. Access Cost Explorer and create budgets.
- 5. Purchase and manage Savings Plans.
- 6. Publish [AWS Cost and Usage Reports](#).

Consolidated billing

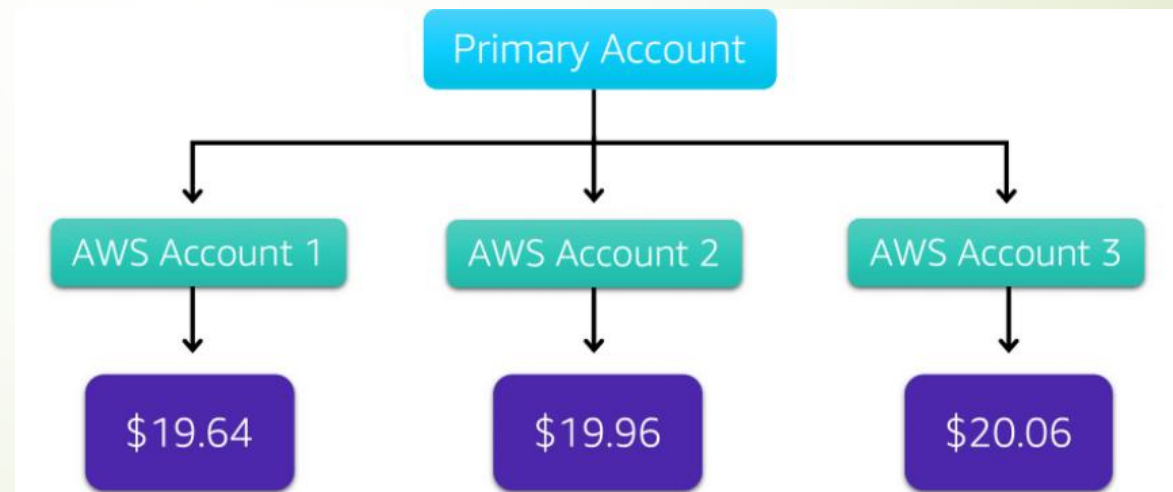
- In an earlier module (Week 7), you learned about AWS Organizations, a service that enables you to manage multiple AWS accounts from a central location. AWS Organizations also provides the option for consolidated billing.
- The consolidated billing feature of AWS Organizations enables you to receive a single bill for all AWS accounts in your organization. By consolidating, you can easily track the combined costs of all the linked accounts in your organization. The default maximum number of accounts allowed for an organization is **4**, but you can contact AWS Support to increase your quota, if needed.
- On your monthly bill, you can review itemized charges incurred by each account. This enables you to have greater transparency into your organization's accounts while still maintaining the convenience of receiving a single monthly bill.

Consolidated billing

- Another benefit of consolidated billing is the ability to share bulk discount pricing, Savings Plans, and Reserved Instances across the accounts in your organization.
- For instance, one account might not have enough monthly usage to qualify for discount pricing. However, when multiple accounts are combined, their aggregated usage may result in a benefit that applies across all accounts in the organization.

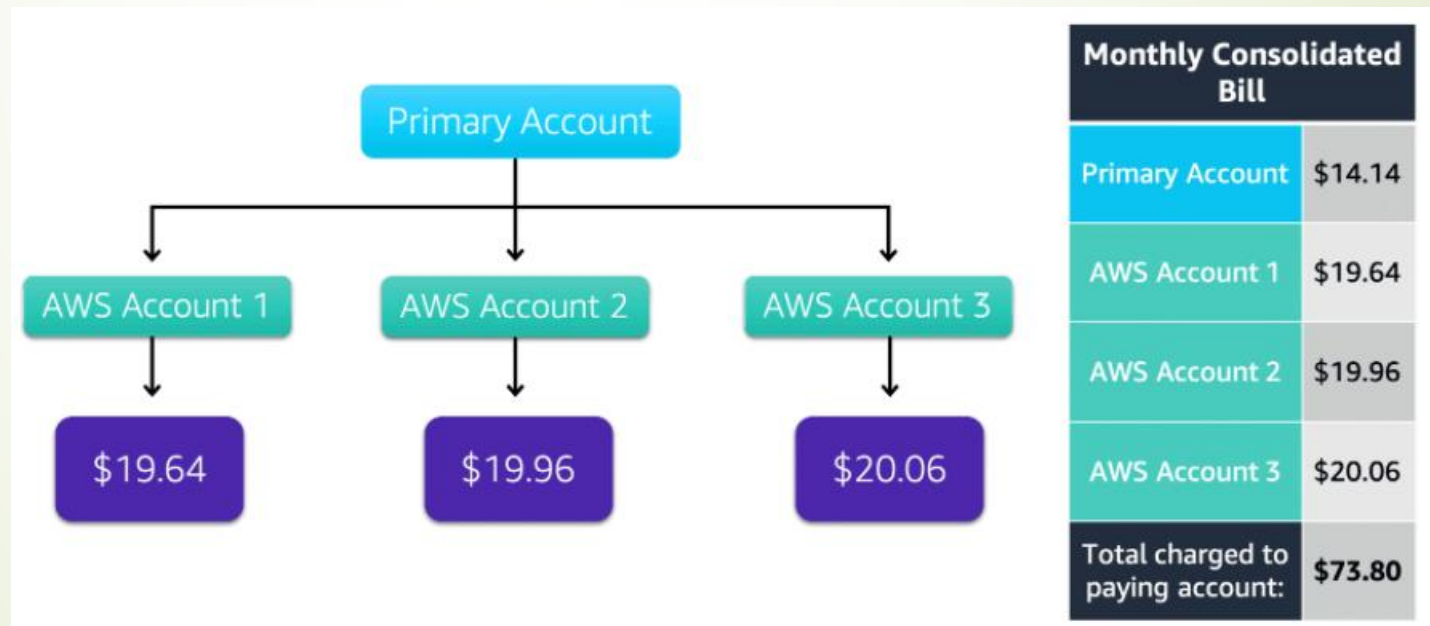
Example: Consolidated billing

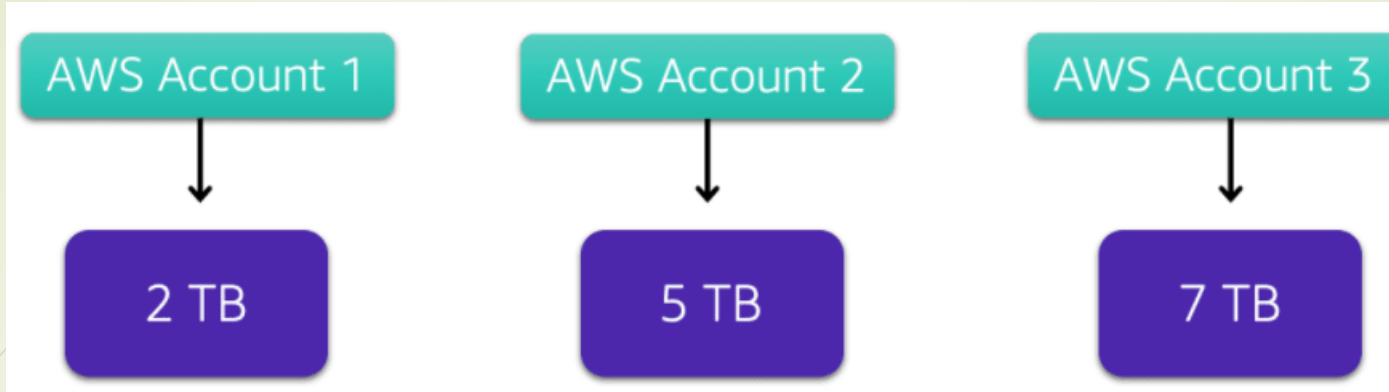
- Suppose that you are the business leader who oversees your company's AWS billing.
- Your company has three AWS accounts used for separate departments. Instead of paying each location's monthly bill separately, you decide to create an organization and add the three accounts.
- You manage the organization through the primary account.



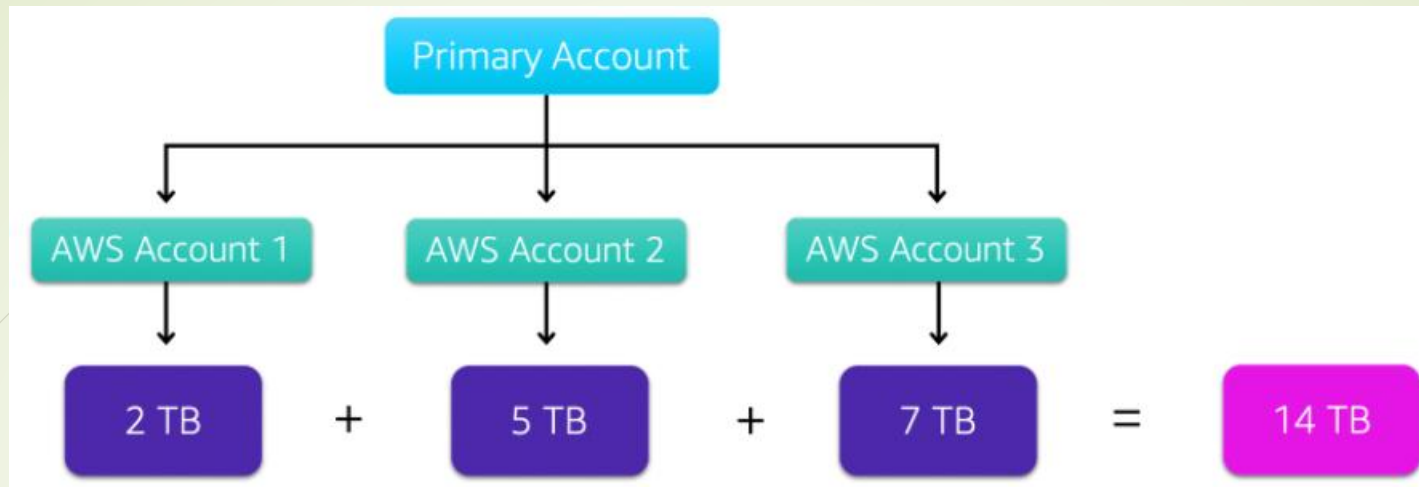
Example: Consolidated billing

- Each month, AWS charges your primary payer account for all the linked accounts in a consolidated bill. Through the primary account, you can also get a detailed cost report for each linked account.
- The monthly consolidated bill also includes the account usage costs incurred by the primary account. This cost is **not a premium charge for having a primary account**.





- Consolidated billing also enables you to share volume pricing discounts across accounts.
- Some AWS services, such as Amazon S3, provide volume pricing discounts that give you lower prices the more that you use the service.
- In Amazon S3, after customers have transferred **10 TB of data in a month, they pay a lower per-GB transfer price for the next 40 TB of data transferred.**
- In this example, there are three separate AWS accounts that have transferred different amounts of data in Amazon S3 during the current month:
 - Account 1 has transferred 2 TB of data.
 - Account 2 has transferred 5 TB of data.
 - Account 3 has transferred 7 TB of data.
- Because no single account has passed the 10 TB threshold, none of them is eligible for the lower per-GB transfer price for the next 40 TB of data transferred.



- Now, suppose that these three separate accounts are brought together as linked accounts within a single AWS organization and are using consolidated billing.
- When the Amazon S3 usage for the three linked accounts is combined (2+5+7), this results in a combined data transfer amount of 14 TB. This exceeds the 10-TB threshold.
- With consolidated billing, AWS combines the usage from all accounts to determine which volume pricing tiers to apply, giving you a lower overall price whenever possible. AWS then allocates each linked account a portion of the overall volume discount based on the account's usage.
- In this example, Account 3 would receive a greater portion of the overall volume discount because at 7 TB, it has transferred more data than Account 1 (at 2 TB) and Account 2 (at 5 TB).

AWS Budgets

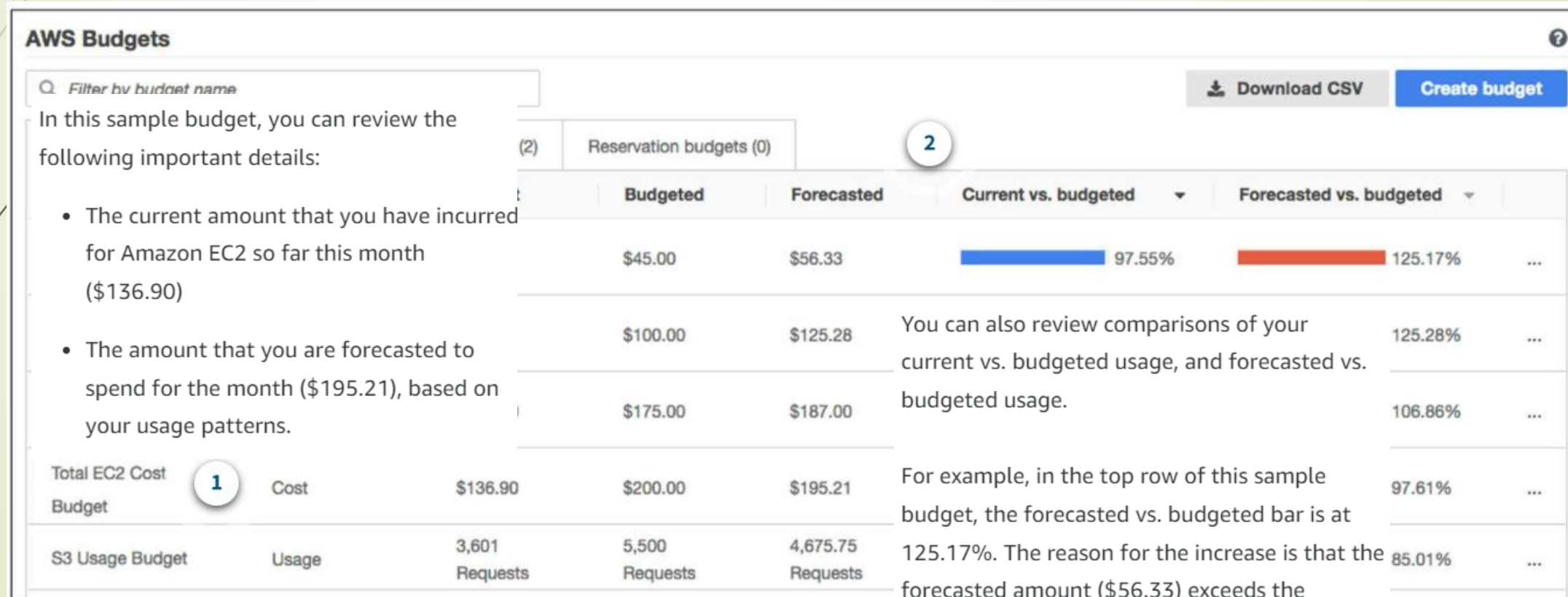
AWS Budgets

- In [AWS Budgets](#), you can create budgets to plan your service usage, service costs, and instance reservations.
- The information in AWS Budgets updates three times a day. This helps you to accurately determine how close your usage is to your budgeted amounts or to the AWS Free Tier limits.
- In AWS Budgets, you can also set custom alerts when your usage exceeds (or is forecasted to exceed) the budgeted amount.

Example: AWS Budgets

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- Suppose that you have set a budget for Amazon EC2. You want to ensure that your company's usage of Amazon EC2 does not exceed \$200 for the month.
- In AWS Budgets, you could set a custom budget to notify you when your usage has reached half of this amount (\$100). This setting would allow you to receive an alert and decide how you would like to proceed with your continued use of Amazon EC2.

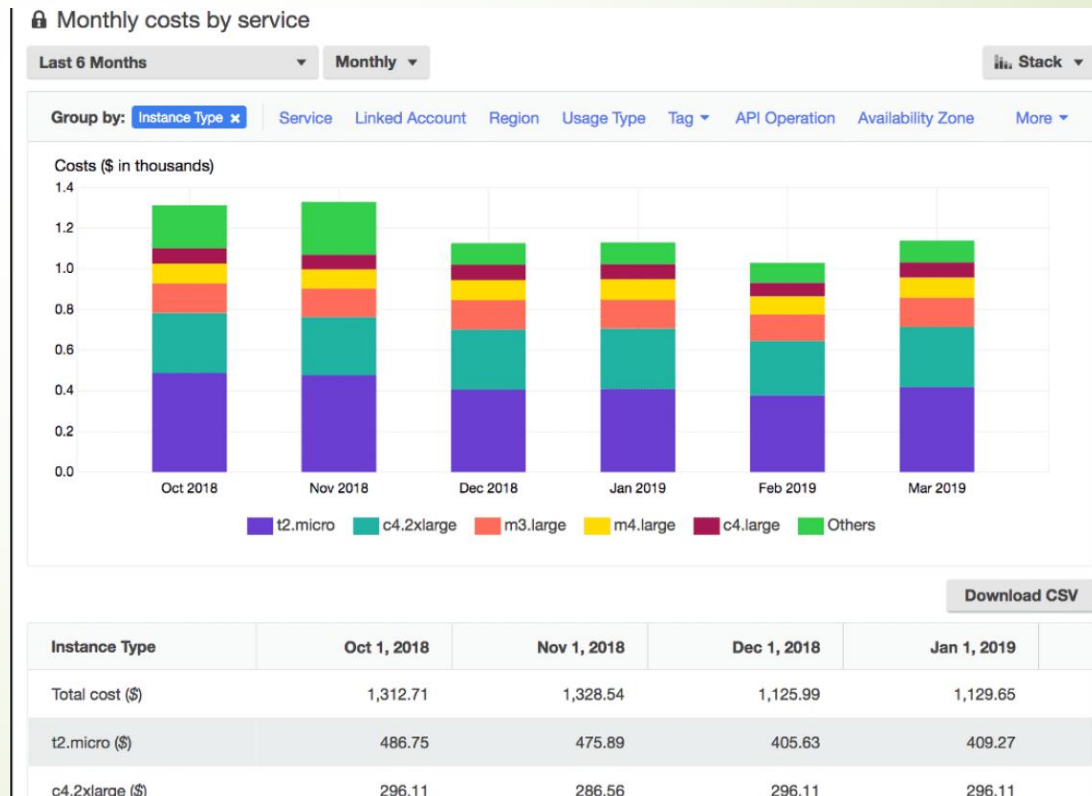


AWS Cost Explorer

- AWS Cost Explorer is a tool that enables you to visualize, understand, and manage your AWS costs and usage over time.
- AWS Cost Explorer includes a **default report of the costs and usage for your top five cost-accruing AWS services**. You can apply custom filters and groups to analyze your data. For example, you can view resource usage at the hourly level.
- It's a console-based service that allows you to visually see and analyze how you are spending money with AWS. It will show you which services you are spending the most money on, and it gives you **12 months of historical data**, so you can track your spending over time.
- That way, if you see a bump in spending on, say, EC2 from October to December, you can then use that data to go on and figure out why exactly that happened.

Example: AWS Cost Explorer

- This example of the AWS Cost Explorer dashboard displays monthly costs for Amazon EC2 instances over a 6-month period. The bar for each month separates the costs for different Amazon EC2 instance types (such as t2.micro or m3.large).
- By analyzing your AWS costs over time, you can make informed decisions about future costs and how to plan your budgets



AWS Cost Explorer is highly customizable

- Cost Explorer gives you some powerful defaults for reports, but you can build your own custom ones as well. This will help you identify cost drivers and take action when necessary to curb spending. Cost optimization is a priority you should be paying close attention to, and you can use the Cost Explorer to help get you going in the right direction.
- You can change the timeframe to specific number of months and change the data to be grouped by different attributes such Region or group by other attributes as well.
- One important grouping to note is to group by tag. Many resources in AWS are taggable. Tags are essentially **user-defined key-value pairs**. So you can tag an EC2 instance with a specific project name or a database with the same project name, and then you can come into the AWS Cost Explorer, filter by tag, and see all of the expenses associated with that tag.
- Cost Explorer also allows you to create custom reports, e.g. for the daily cost for the month of January of this year, you can then group by service. You can also save a report and come back to it when it is needed.



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AWS Support plans

AWS Support plans

- AWS offers four different Support plans to help you troubleshoot issues, lower costs, and efficiently use AWS services.
- It is best practice to choose a Support plan to meet your company's needs:
 - Basic
 - Developer
 - Business
 - Enterprise

Basic Support

- **Basic Support** is free for all AWS customers. It includes access to whitepapers, documentation, and support communities. With Basic Support, you can also contact AWS for billing questions and service limit increases.
- With Basic Support, you have access to a limited selection of AWS Trusted Advisor checks.
- Additionally, you can use the **AWS Personal Health Dashboard**, a tool that provides alerts and remediation guidance when AWS is experiencing events that may affect you.
- If your company needs support beyond the Basic level, you could consider purchasing Developer, Business, or Enterprise Support.

Developer, Business, and Enterprise Support

- The Developer, Business, and Enterprise Support plans include all the benefits of Basic Support, in addition to **the ability to open an unrestricted number of technical support cases**.
- These three Support plans have **pay-by-the-month pricing** and require **no long-term contracts**.
- The information in this course highlights only a selection of details for each Support plan. A complete overview of what is included in each Support plan, including pricing for each plan, is available on the [AWS Support](#) site.
- In general, for pricing, the Developer plan has the lowest cost, the Business plan is in the middle, and the Enterprise plan has the highest cost.

Developer Support

- Customers in the **Developer Support** plan have access to features such as:
 - Best practice guidance
 - Client-side diagnostic tools
 - Building-block architecture support, which consists of guidance for how to use AWS offerings, features, and services together
- For example, suppose that your company is exploring AWS services. You've heard about a few different AWS services. However, you're unsure of how to potentially use them together to build applications that can address your company's needs.
- In this scenario, the building-block architecture support that is included with the Developer Support plan could help you to identify opportunities for combining specific services and features.

Business Support

- ▶ Customers with a **Business Support** plan have access to additional features, including:
 - ▶ Use-case guidance to identify AWS offerings, features, and services that can best support your specific needs
 - ▶ All AWS Trusted Advisor checks
 - ▶ Limited support for third-party software, such as common operating systems and application stack components
- ▶ Suppose that your company has the Business Support plan and wants to install a common third-party operating system onto your Amazon EC2 instances. You could contact AWS Support for assistance with installing, configuring, and troubleshooting the operating system. For advanced topics such as optimizing performance, using custom scripts, or resolving security issues, you may need to contact the third-party software provider directly.

Enterprise Support

- In addition to all the features included in the Basic, Developer, and Business Support plans, customers with an **Enterprise Support** plan have access to features such as:
 - **Application architecture guidance**, which is a consultative relationship to support your company's specific use cases and applications
 - **Infrastructure event management**: A short-term engagement with AWS Support that helps your company gain a better understanding of your use cases. This also provides your company with architectural and scaling guidance.
 - A Technical Account Manager

Technical Account Manager (TAM)

- The TAM is the primary point of contact at AWS for companies that subscribe for Enterprise Support. They provide guidance, architectural reviews, and ongoing communication with your company as you plan, deploy, and optimize your applications.
- Your TAM provides expertise across the full range of AWS services. They help you design solutions that efficiently use multiple services together through an integrated approach.
- For example, suppose that you are interested in developing an application that uses several AWS services together. Your TAM could provide insights into how to best use the services together. They achieve this, while aligning with the specific needs that your company is hoping to address through the new application.

Knowledge check

- Which Support plan includes all AWS Trusted Advisor checks at the lowest cost?
 - A. Basic
 - B. Developer
 - C. Business
 - D. Enterprise
- C
- Only the Business and Enterprise Support plans include all AWS Trusted Advisor checks. Of these two Support plans, the Business Support plan has a lower cost.

AWS Marketplace

AWS Marketplace

- [AWS Marketplace](#) is a digital catalog that includes thousands of software listings from independent software vendors. You can use AWS Marketplace to find, test, and buy software that runs on AWS.
- For each listing in AWS Marketplace, you can access detailed information on pricing options, available support, and reviews from other AWS customers.
- You can also explore software solutions by industry and use case.
- For example, suppose that your company is in the healthcare industry. In AWS Marketplace, you can review use cases that software helps you to address, such as implementing solutions to protect patient records or using machine learning models to analyze a patient's medical history and predict possible health risks.

AWS Marketplace categories

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- AWS Marketplace offers products in several categories, such as **Infrastructure Products**, **Business Applications**, **Data Products**, and **DevOps**.
- Within each category, you can narrow your search by browsing through product listings in subcategories. For example, subcategories in the DevOps category include areas such as **Application Development**, **Monitoring**, and **Testing**.



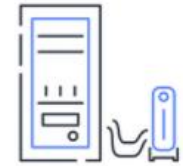
Business Applications



Data & Analytics



DevOps



Infrastructure Software



Internet of Things
(IoT)



Machine Learning



Migration



Security

AWS Marketplace – Features & Benefits

- The AWS Marketplace is a curated digital catalog that streamlines your steps to find, deploy and manage third party software running in your AWS architecture, all while providing a range of payment options, and this allows you to accelerate innovation while rapidly and securely deploying a wide range of solutions, while also reducing your total cost of ownership.
- The first key way that the AWS Marketplace helps customers, is that instead of needing to build, install and maintain the foundational infrastructure needed to run these third party applications in the marketplace, customers have options like **one-click deployment** that allows them to quickly procure and use products from thousands of software sellers right when you need them.

AWS Marketplace – Features & Benefits

- Every vendor can choose what options are available, but almost every vendor in the marketplace will **allow you to use any annual licenses you already own and credit them for AWS deployment**
- As you move forward, most vendors in the marketplace also offer **on-demand pay-as-you-go options**. These flexible pricing plans give you more options to pay for the software, the way you actually use it without wasted unused licenses weighing down your balance sheets.
- Many vendors even offer **free trials or Quick Start plans** to help you experiment and learn about their offerings.
- The marketplace offers a number of enterprise-focused features, such as custom terms and pricing, where you can manage custom licensing agreements. A private marketplace, where you can create a custom catalog of pre-approved software solutions that meet your specific legal or security standards.

Summary

- ▶ In this Module, you learned about the following concepts:
 - Three types of offers included in the AWS Free Tier: 12 months free, Always free, and Trials
 - Benefits of consolidated billing in AWS Organizations
 - Tools for planning, estimating, and reviewing AWS costs
 - Differences between the four AWS Support plans: Basic, Developer, Business, and Enterprise
 - How to discover software in AWS Marketplace

Additional resources

- To learn more about the concepts that were explored in this Module, review these resources.
- [AWS Pricing](#)
- [AWS Free Tier](#)
- [AWS Cost Management](#)
- [Whitepaper: How AWS Pricing Works](#)
- [Whitepaper: Introduction to AWS Economics](#)
- [AWS Support](#)
- [AWS Knowledge Center](#)